

08/01/25

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Software Engineering and Project Management

Software - A collection of programs along with documentation associated.

Software Engineering - involves production of software. It is an engineering discipline that includes all aspects of software production.

Software process - Process of developing software product.

Developing software:

↳ Requirement specification

↳ Generic: with developer

↳ Customised: by user.

* Maintenance is key in software development

need for S.E. ∴ Demand ↑ Production ↑
Cost → Development 60%
→ Maintenance 40% (60%)
40% → for complex systems

Essential attributes of good software:

- Maintainability
- Dependability
- Efficiency
- Acceptability

Maintainability - addressing changing needs of a customer.

Dependability = Reliability and Trustworthy

Safety and Security of the software as key.

Efficiency ~ Memory, processor cycle

Acceptability ~ appreciated by the user.
meets all the requirements.

Model - Abstraction of the software process.

Diff between S.E. & C.S

SE is subset of

system design development

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Ethical Responsibility

- Not to misuse resources of the organisation
- They should uphold the law.
- Faithful to the organisation.

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Software Process Activities:

- Specification
- Design & Implementation.
- Validation
- Evolution.

5/3

Specification: function required by a software.

Design: ~~As~~ Making Architectural model.

Represented using UML diagrams.

Validation: Checking & Verifying if the developed product meet the needs.

Evolution: New requirements may emerge upon using.

slow approach
dev

1. Plan Driven
2. Agile Approach

Approach

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Process Models:

- 1] Waterfall model
- 2] Incremental S/W model
- 3] Reuse Oriented S/W dev. model.

1] Waterfall Model: 5 Phases; Plan Driven

→ Utilised when Requirement Analysis is well understood.

→ Unchanging R.A.

→ Security is important then use this

2] Incremental Model: Agile

→ Requirement Specification & Design are done simultaneously

→ Incrementally dev.

→ R.A. need not be done completely

→ Changing Req. Specification

→ Used in GUI.

Complex
Security
concerns

They will give
as a case study on type of model
house.

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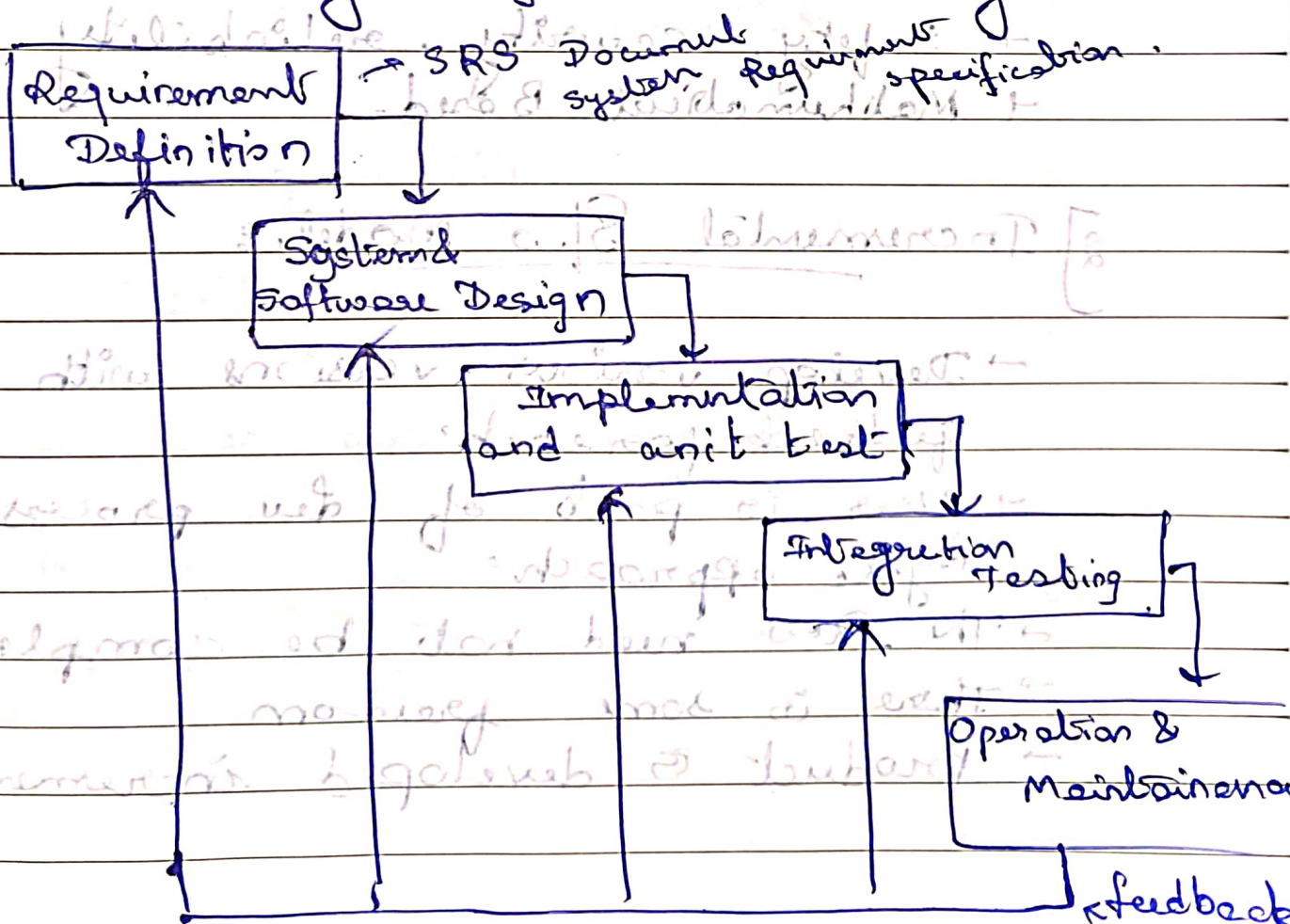
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- 3] Reuse Oriented in design of system
- Existing Software can be built on upon
 - No need to create from scratch.

1] Water fall Model -

- Based on plan-driven approach.
- Requirement Specification must be ready to go to next stage.



Note | User is part of development process - Agile. -

Dis Adv :

- Any changes to be made must wait till the end.

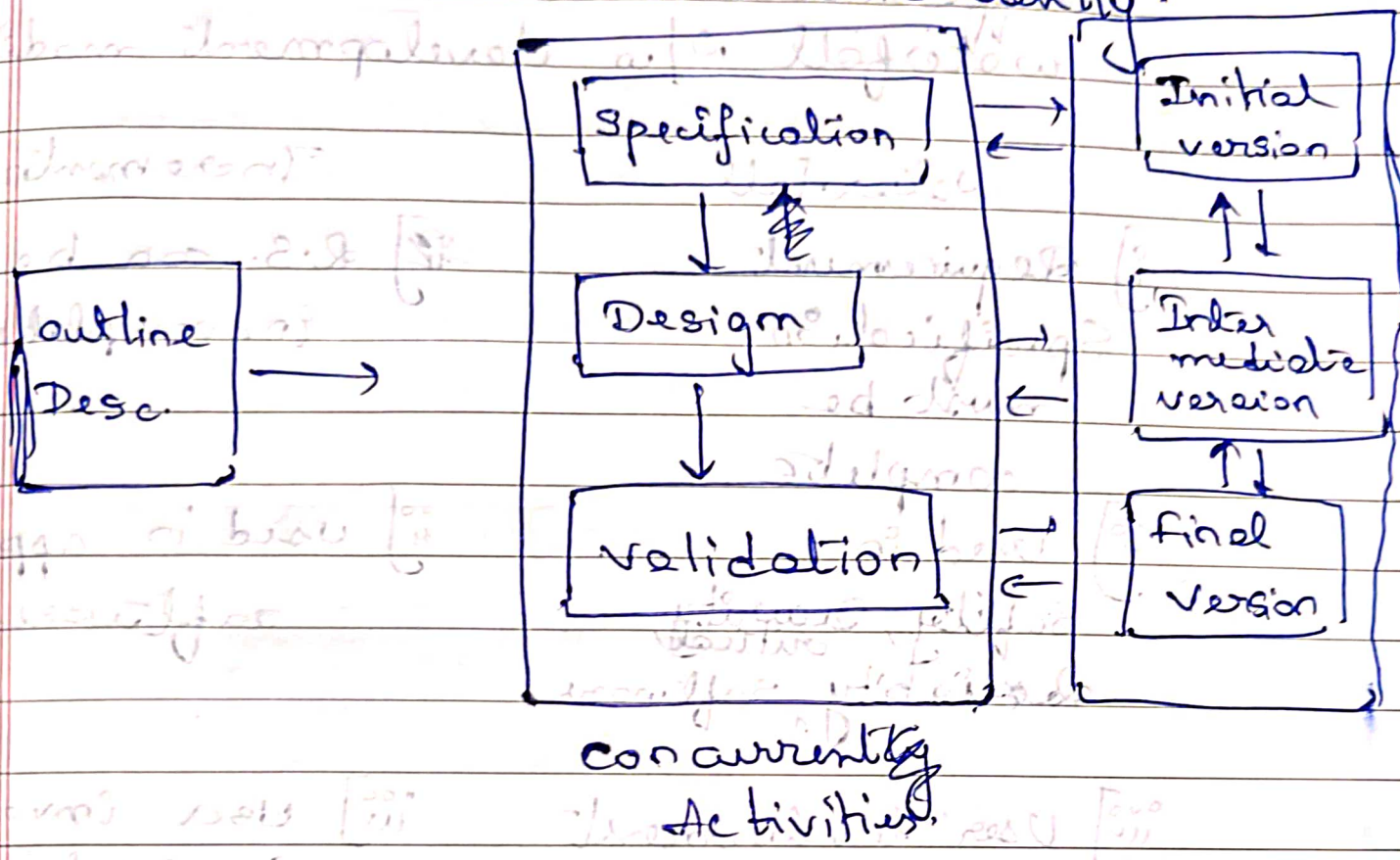
Formal System Development (waterfall model)

- variant of waterfall
- Based on mathematical models
- Safety, security, reliability
- Mathematical Based

2] Incremental Spw Model :

- Develop various versions with user feedback for each
- User is part of dev. process.
- Agile approach.
- The R.O.S. need not be complete
- There is some freedom
- Product is developed incrementally.

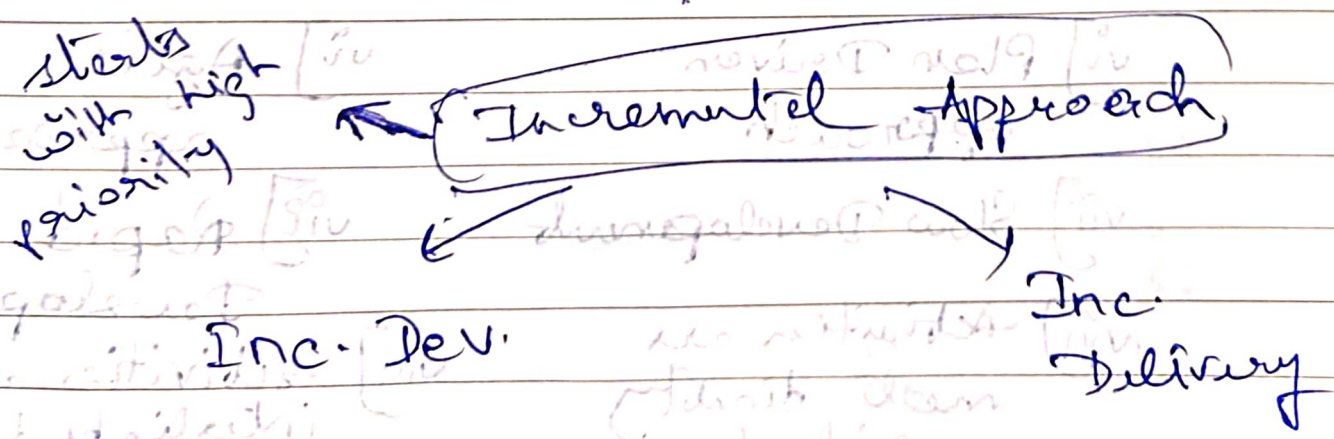
→ All activities are done concurrently.



- Ex: Developing UI.

Application System

→ Less cost ^{then} and waterfall.



Difference between Incremental and waterfall S/W development models

waterfall

- i] Requirement Specification must be complete
- ii] Used in Safety, Security, Reliability, ^{critical} software.
- iii] User involvement is low
- iv] Any changes need to wait till end of the cycle.
- v] Cost is High
- vi] Plan Driven approach
- vii] Slow Development
- viii] Activities are mostly directly

Incremental

- i] R.S. can be incomplete
- ii] Used in application software
- iii] User involvement is high.
- iv] Changes can be incorporated concurrently
- v] Cost Effective
- vi] Agile approach
- vii] Rapid Development
- viii] Activities are interlinked

Computer Science connected with theories and methods that underlie computers

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3] Reuse Oriented Development Model -

→ Using already existing software.

Adv.

→ Cost of dev is saved.

→ We know if user likes it or not.

Assignment Questions:

need and

1] Explain Importance of Studying S.E.?

2] Define i] Software ii] S.E.

3] List out Diff. btw

i] S.E. and C.S. ii] S.E. and System Eng.

4] Explain the essential attributes of good Software

5] Briefly explain about costs of SE and key challenges of SE.

6] List and explain Software Process activities

7] Define Software process and software process model

8] List out the diff between generic and customised products.

9] Briefly explain Plan-Driven and agile process (Diff).

system engineering → development
and evaluation of complex systems

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- 10] Explain S.E. ethics and issues of professional responsibility with code of ethics defined by ACM / IEEE.
- 11] Explain water fall model along with the diagram. List adv and dis adv.
- 12] ~~12]~~ Explain Incremental model along with the diagram. List adv and Dis adv.

Disadv of Incremental:

Process

- User is part of development. So less security & confidentiality.
- Constantly new version are being updated. So users have to update frequently.
- Document keeps on changing.

Adv. of Incremental:

- User is part of development. So we know about the result before deployment.
- Cost Efficient.

Challenges of S.E:

- High learning Curve.
- Maintenance can be hectic.
- Preparation of documentation can be difficult.

13] why 'one approach can't be used for large as well as complex system?

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(A) Spiral Model → Changes to be made are given during dev. So, incorporating them can be difficult.

→ Activities are interlinked. So it is difficult to break down ^{large system} into smaller parts.

→ Documents will also be complex. So can't edit it multiple times.

→ Complex system needs clear understanding of R.A. So it is important to know them well in advance.

→ Final product needs rigorous testing. It is not explicitly mentioned in incremental model.

Boehm

Boehm's Spiral Model:

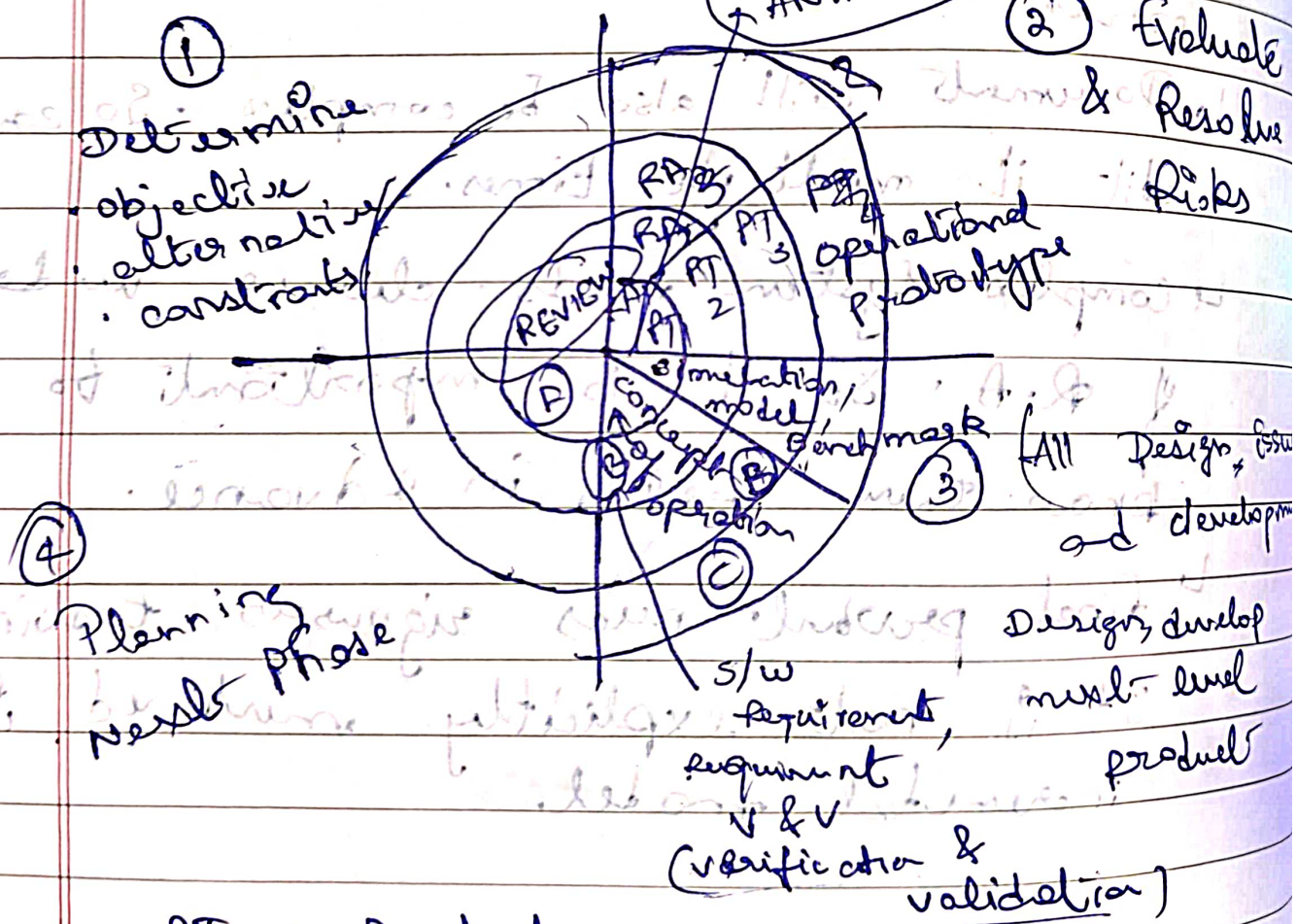
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- 4 Phases
- Spiral fashion

4 spiral

(Incremental approach)

- Objectives, alternatives & Constraints are noted.



PT → Prototype

RA → Risk analysis

③ Design . Design V&V

④ ~~Very~~ Detailed Design - Coding ~~that~~
 Unit Test, Integration Test,
 Acceptance Test (user level).
~~Int~~

⑤ Requirement plan
 Life cycle Plan
 Development Plan
 Integrative Test Plan

all in each spiral.

→ Spiral Model is used when there is risk involved in the project.

Risk: i) Using new programming language.

ii) Unethical developers

iii) Incompetent developers.

Risk management is part of Software project management.

Spiral Model:

- | | | |
|------|---------------------------|------------------------|
| I] | Objective Setting | } All are interrelated |
| II] | Risk Analysis & Reduction | |
| III] | Design and Development | |
| IV] | Planning next phase. | |

→ In Incremental approach, we use the concept of prototyping.

Planning: Req. Plan

Life cycle Plan

Development Plan

Integration Test Plan

→ Suitable for small (or) medium projects.

Objective Setting:

- Objects are set
- Risks are identified
- Finds out the possible alternatives

Object SettingRisk Analysis & Reduction:

- All risks should be analysed, and resolved

Design & Dev

- Choosing appropriate model
- Designing services for the A

Planning next phase:

Initiation is a process of setting up a project. It involves defining the project's purpose, scope, and objectives. It also involves identifying the project's stakeholders and the resources needed to complete the project.

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14] with a neat dig explain Boehm's spiral model.
List out adv & Disadv.

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List out Adv and Disadv of Spiral Model.

Adv :

1] It follows incremental approach
So R.S. need not be complete

Key factors

2] It is a thorough process. So there is a high chance of user acceptance.

3] Risk Management.

Disadv :

1] Very long process

2] Too costly to use in reality

3] A lot of resources are needed.

4] There could be repetition of work.

5] Difficult to track completion progress.

NIMP

Req. Process Engineering is a critical step in S.Dev. why?

System Requirement, User Requirement

2 types of Requirement.

* Feasibility Study is imp in Req. Engineering.

03/02/25

IMP for MSE

water fall
iterative

Basic
Bohem

Ch1. Introduction
Ch2. Process Model
Ch3. Req Engineering
Except Validation

Software Process Activities:

- 1) Software Specification
- 2) Design and Implementation
- 3) Validation
- 4) Evolution

Explain
Software
Process
Activities
w.r.t an
application.

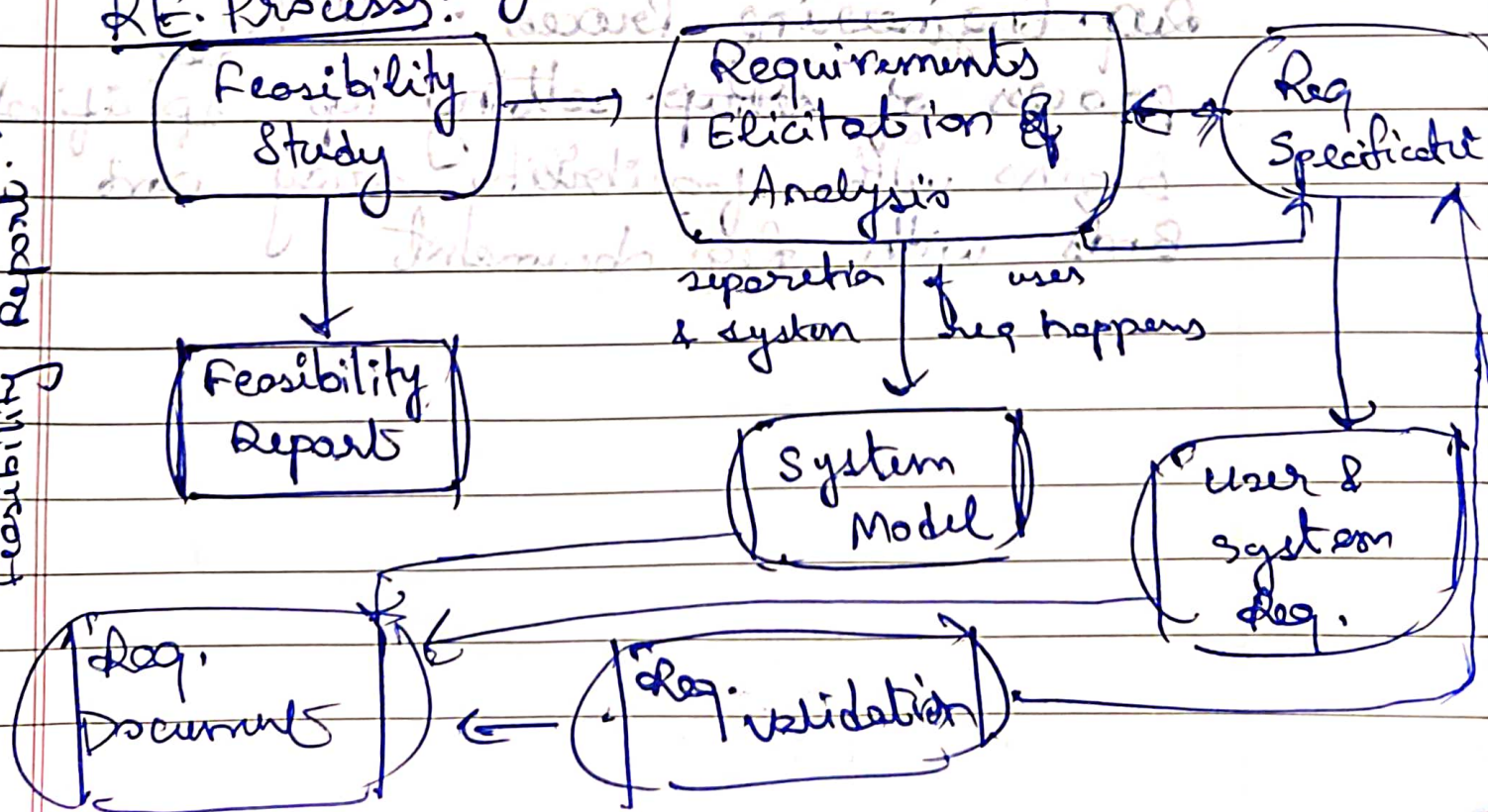
1) Software Specification:

→ Result is Requirement Document: contains all the user requirements and system requirement.

→ Outcome of any process activity is a report.

→ ① Feasibility: Check note and side page.

RE. Process:



is concerned to know about what is available
needed from, sketch, budget, outcome in
Report.
and what is feasibility

Functional service

Feasibility study

Constraints on the operation

are should be clearly known

Note: Requirement Document should contain any error. If there is error in Req. Document, leads to propagation of the errors to the subsequent steps of development and finally leads to increase of the product. That leads to higher maintenance cost. (1) Feasibility study is conducted under this.

(2) Requirements Elicitation of Analysis involves setting up user and system requirements. (decide about a model.)

System Modelling is also an essential part of Requirement specification.

Req. Engineering Process: It is a process of setting up specifications. Begins with feasibility study and ends with Req. document.

To meet user expectations ^{make} system

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Requirement

- Functional: Main Service / Function it has to perform.
- Non-Functional: Secure, Reliable, properties of product

2) S/w Design & Development: Implementation

Design: Deciding the model.

Implementation: Coding

Requirement Document is transformed into Executable S/w System

Design phase: developing architecture of the system.

Components → Subsystems → System

General model of Design process:

Design I/p → Process Activities → Design O/p

System Req → F.R. Users

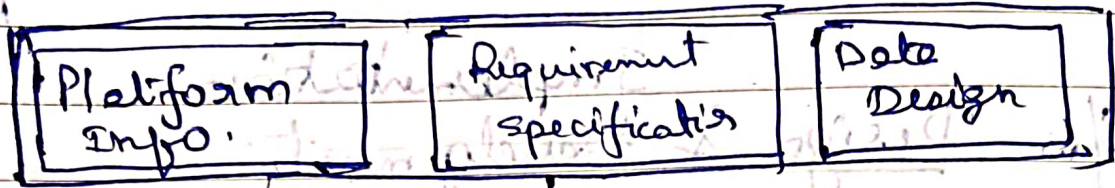
N.F.R. System

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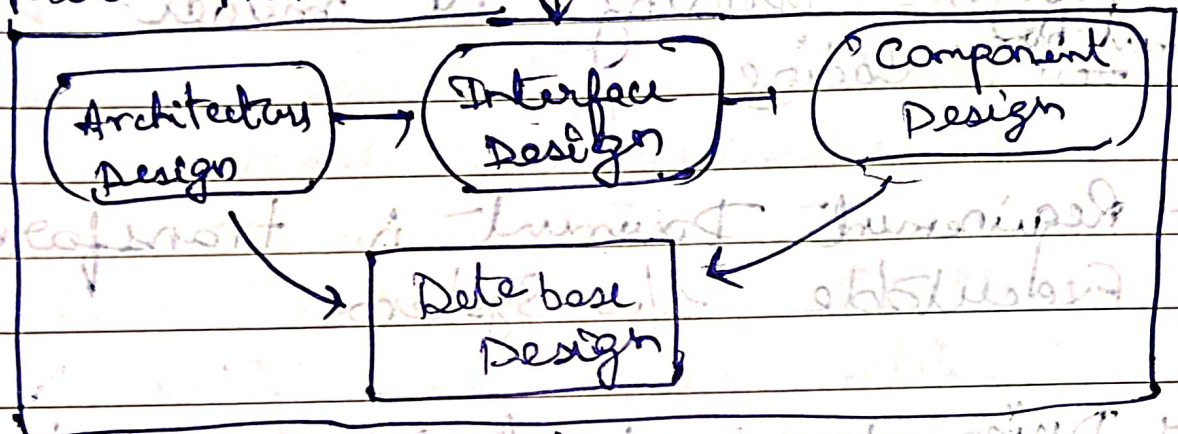
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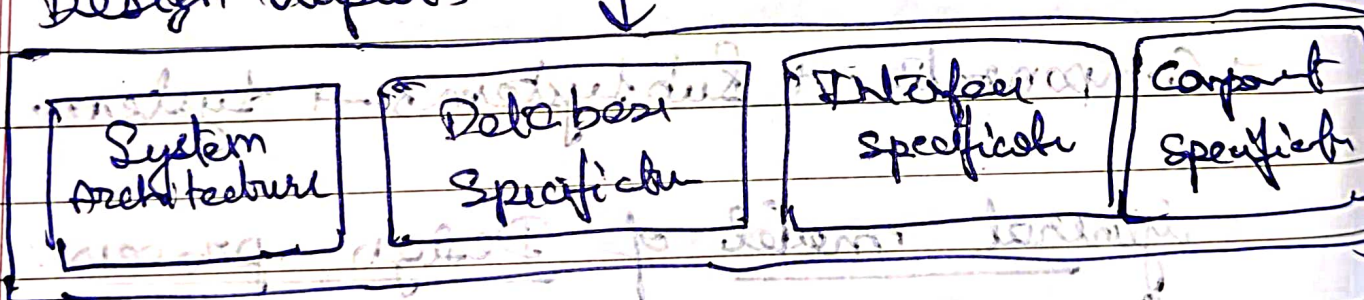
Design Inputs



Process Activities



Design Outputs



* S/w Validation

- 1) Component Testing
- 2) Integration Testing System Testing
- 3) Acceptance Testing & α , β Testing

Req. Document is used by a lot of people in creating product.

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S/w Evolution:

Existing System



user →

Changes



Modification



New System

07/02

Requirements Document has a structure.

- i] Preface
- ii] Introduction
- iii] Glossary
- iv] System Architecture. User Req. Specification
- v] System Architecture → system Requirement
- vi] System Models
- vii] System Evolution
- viii] Appendices
- ix] Index

Functional Req are associated with all the services that are to be provided with system by

Functionality = Services.

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Non-Functional (Non-Emergent System Req)
which includes Reliability, Safety, Maintainability, Availability, Performance

Preface: Gives information about the book. Brief summary of document.

Introduction: Introduce to the problem statement and methodology.

Appendices: Extra information needed.

Indexing: Is a tool to make search operation faster.

Ex: Requirement Document for Hospital Management System. | Patient care

Preface:

Functional Req:

→ Maintain Record of Patient,

like including and track Patient

status from Admitting to Discharge

→

Architecture - Overall design of system

System Req → PR
 → NFR

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Non-Functional Req:

→ Security, Data Privacy

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RE Process

1) Req. Elicitation analysis

2) Req. validation specification → VR

3) Req. validation → SR

Elicitation: A group of people interact together (Stakeholders, etc) collectively come up with Requirement statements.

User Req:

user req
system req

1) Simple natural language

2) Structured natural language - Standard Template

3) Graphical Notations - UML

4) Database Design

(UML - Unified Modelling Language)

5) Mathematical Specification

for formal models

→ Case Study: Group of mental health patients.

→ we need to consider actors of the system.

→ If we are aware of the environment in which system is being developed, it will be more easy.

very difficult as user gives vague answers. (layman terms)

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Elicitation Process has 4 - phases cyclic:

i) Requirement Discovery: Discussion with all people who, part of the development process.

ii) Requirement Classification: grouping similar requirements in one group/ classification: grouped in cluster.

iii) Requirement Prioritization & Negotiation:

iv) Requirement Specification: for documenting

Requirement Validation:

→ To make sure there are no errors.

→ If there are any errors they will be propagated to further levels.

→ Check for completeness.

→ Consistency

→ Test Cases are generated to check validity

→ Check for Realistic needs.

Non-Funds
Req.Product Req.
organisational Req.

External Req. - Ethical, legal.

Only Case Study for us.

MHCPMS core study.

Mental Health Care - Patient Management system

IMP

Sequence impacted

Write the scenario of MHC-PMS:

→ Different Actors

→ How system should work for each actor.

→ What can go wrong?

Receptionist:

→ Check Appointment Schedule.

→ Access to edit; update and change schedule.

→ Access to emergency services.

• Patient may deny to provide details

• Patient may give fake details.

Nursing Staff:

→ Access to view, change, update medical history

→ Access to contact doctors for w.

• Patients may deny (or) give fake details.

• Patient not available in database.

Doctor:

→ Access to medical history of patients

→ Update new reports of patients

→ Make prescriptions.

• No Patient records available.

Remember only for Mental Health.

Patient:

- Book appointment.
- Check prescription, doctor details
- Check bills.
- Misinformation in the system.

Non-Functional Requirements

Product Requirements

- ↳ Usability
- ↳ Efficiency Req.
- ↳ Dependability Req.
- ↳ Security Req.
- ↳ Efficiency Req.
- ↳ Performance Req.
- ↳ Space Req.

Organizational Requirements

- ↳ Environmental Req.
- ↳ Operational Req.
- ↳ Development Req.

External Requirements

- ↳ Regulatory req.
- ↳ Ethical Req.
- ↳ Legislative Req.

Accounting Req.

Safety/Security requirements